William Hogg

Project Portfolio:
WilliamWHogg.com

E-mail: WilliamWHogg@gmail.com **Phone:** (206) 953-7510 **LinkedIn:** linkedin.com/in/will-hogg

Education

M.S. in Mechanical Engineering - GPA: 4.0 ***May 2026***
Washington State University, Pullman, WA
Relevant Coursework: Additive Manufacturing, Cryogenics, Machine Learning, Topology Optimization, Robotics

B.S. in Mechanical Engineering - GPA: 4.0 ***May 2025***
Washington State University, Pullman, WA
Minors: Material Science, Mathematics
Relevant Coursework: Mechanics of Solids, Mechatronics, Material Characterization, Linear Optimization

Experience

WSU - Thermal and Fluids Laboratory – Instructor Assistant ***Jan 2026 – May 2026***
- Guided junior-level students in experimental setup, uncertainty analysis, and data interpretation for thermal–fluids experiments

W.M. Keck Biomedical Materials Lab – Research Assistant ***Aug 2025 – Dec 2025***
- Optimized L-PBF printing of brittle MMCs using batch Bayesian optimization, reducing cracking and enabling successful builds on a 3DS DMP 200 system
- Developed custom MATLAB apps to automate powder particle analysis and mechanical test data processing, reducing analysis time by ~5×
- Maintained, repaired and trained others on metal laser powder bed fusion (LPBF) systems

W.M. Keck Biomedical Materials Lab - Undergraduate Researcher ***Apr 2023 – Aug 2025***
- Developed a novel protocol for corrosion-fatigue testing of LPBF alloys, resulting in a first-author publication
- Programmed CAM (Fusion 360) for CNC milling and turning of miniaturized hip joint components to evaluate novel alloys
- Performed comprehensive materials characterization (tensile, hardness, optical microscopy, SEM, fractography) on novel additively manufactured alloys

Exotic Metals Forming – Project Engineer Intern ***Summer 2022***
- Conducted root cause analysis on titanium superplastic forming (SPF) parts, preventing failures and reducing tool wear
- Updated legacy GD&T documentation, reducing operator confusion and rework in manufacturing processes
- Designed and implemented an ergonomic tool for roller seam welding of a 90 lb component

Frank Innovation Zone (FIZ) - Instructor/Technician ***Jan 2022 – Apr 2023***
- Trained 100+ students and researchers in safe use of shop equipment
- Maintained a 3,000+ sq ft maker space, while securing $80k of new equipment

Eagle Scout – Troop 624 ***2014 – 2021***
- Led 150+ hour eagle scout project during the covid pandemic

Featured Projects – Details on WilliamWHogg.com

Acoustic and Thermal Monitoring of Wire Arc Directed Energy Deposition	***Dec 2025***
Design and Validation of a Classroom-Scale Injection Molding Machine	***Apr 2025***
Deployable Multi-Actuator Wheelchair Lift	***Dec 2024***

Skills & Abilities

Design: SOLIDWORKS, 3DXpert, GD&T, Fusion 360, Siemens NX, COMSOL, MATLAB, Python, Arduino
Manufacturing: Additive Manufacturing (FDM, SLA, LPBF, DED), CNC Machining, Sheet Metal Forming
Characterization: Mechanical Testing (Fatigue, Tension/Compression, Hardness), Optical Microscopy, SEM/EDS Analysis

Publications and Honors

"Corrosion-Fatigue of Additively Manufactured Ti6Al4V" *JMBBM*	***Dec 2025***
VCEA-Outstanding Junior in Mechanical Engineering	***April 2024***
"Fatigue Behavior of Additively Manufactured Ti3Al2V Alloy," *MSAM*	***Sep 2023***